

Cell Design Specification — V4b "h60 + anode margin"

Document number	VBF-T1R2-DS-01
Case	t1_r2 — next-generation pure electric sedan cell
Design target	ED $\geq$ 392.61 Wh/kg; 4C fast charge, no lithium plating; $T_{\text{max}} \leq 60$ ° C (333.15 K); overcharge to 4.7 V without thermal runaway
Nominated design	V4b (Round 4) — all mechanical criteria pass
Base parameter set	Chen2020 (PyBaMM, NMC811 / graphite)

1. Basic specification

Item	Value	Source
Electrochemical system	NMC811 positive / graphite negative, LiPF ₆ in EC/EMC	base parameter set Chen2020
Nominal capacity	5.0 Ah	parameter set 'Nominal cell capacity [A.h]'
Simulated 1C discharge capacity	5.6997 Ah	'cell/r4_v4b_1c_spme.json:capacity_ah'
Voltage window	2.5 – 4.2 V	parameter set 'Lower/Upper voltage cut-off [V]'
Electrode stack thickness	194.6 μm (75.6 + 96 + 9 + 8 + 6)	'cell/r4_v4b_energy.json:thickness_m' (mechanical sum of layer thicknesses)
Electrode area	0.1027 m ²	'cell/r4_v4b_energy.json:area_m2'
Cell height $\times$ width	Not provided (cell-can form factor not modeled)	—
Design cell volume (thermal sim)	2.514e-5 m ³	'cell/params_v4b.json:Cell volume [m3]' (scaled 2.42e-5 $\times$ 208.6/200.8; corresponds to the pre-collector-thinning stack — conservative for temperature, see § 6 note)
Electrolyte formulation	$\sigma = 1.8 \text{ S/m}$; $D = 6.0\text{e-}10 \text{ m}^2/\text{s}$; $t_+ = 0.4$ (base 0.2594)	bridge values: literature-based estimates, marked 'estimate' (recorded in propose R2 – R4)
Additive candidates	None proposed (case started at Stage 3; molecular funnel skipped, entry-0 meta)	log.jsonl entry 0
Cation transference number	0.4	'cell/params_v4b.json'

2. Electrode and separator

Layer	Thickness	Porosity	Active material / collector	Particle radius	Source
Positive electrode	75.6 μm	0.36	NMC811	2.0 μm	thickness = parameter set 'Positive electrode thickness [m]'; porosity/particle 'cell/params_v4b.json'
Negative electrode	96 μm	0.32	Graphite	1.5 μm	'cell/params_v4b.json'
Separator	9 μm	0.50	polyolefin (density 397 kg/m ³ , parameter set)	—	'cell/params_v4b.json'
Positive current collector	8 μm	—	Al (2700 kg/m ³ , parameter set)	—	'cell/params_v4b.json'
Negative current collector	6 μm	—	Cu (8960 kg/m ³ , parameter set)	—	'cell/params_v4b.json'

N/P ratio ≈ 1.35 — formula: $N/P = (\text{negative capacity density} \times \text{thickness}) \div (\text{positive capacity density} \times \text{thickness})$.

Mechanical inputs: negative areal mass $108.17 \text{ g/m}^2 \times C_{\text{neg}} = 372 \text{ mAh/g}$ (graphite theoretical, literature); positive areal mass $157.83 \text{ g/m}^2 \times C_{\text{pos}} = 190 \text{ mAh/g}$ (NMC811 practical at 4.2 V, literature range 180 – 200 mAh/g, midpoint 'estimate') $\rightarrow 40.24 / 29.99 = 1.34$; with C_{pos} in range $\rightarrow 1.31 - 1.38$, quoted as ≈ 1.35 (consistent with propose R4). Areal masses: 'cell/r4_v4b_energy.json:layer_kg_m2'.

3. Process design parameters

Parameter	Value	Formula	Source note
Positive areal density	157.83 g/m^2	$\text{thickness} \times (1 - \text{porosity}) \times \text{electrode density}$	'r4_v4b_energy.json:layer_kg_m2.positive_electrode' ($75.6\text{e-}6 \times 0.64 \times 3262$)
Negative areal density	108.17 g/m^2	same	'layer_kg_m2.negative_electrode' ($96\text{e-}6 \times 0.68 \times 1657$)
Positive compaction density	2.09 g/cm^3	$\text{electrode density} \times (1 - \text{porosity}) \div 1000$	$3262 \times 0.64 = 2087.7 \text{ kg/m}^3$; densities = parameter set
Negative compaction density	1.13 g/cm^3	same	$1657 \times 0.68 = 1126.8 \text{ kg/m}^3$
Electrolyte fill amount	7.69 g/cell	$\text{pore volume} \times \text{electrolyte density} \times \text{fill factor}$	pore volume $6.41 \text{ cm}^3 = (75.6 \times 0.36 + 9 \times 0.5 + 96 \times 0.32) \mu\text{m} \times 0.1027 \text{ m}^2$; electrolyte density 1.2 g/cm^3 (literature, 'estimate'); fill factor 1.0 assumed
Formation recommendation	0.1C CC to 4.2 V, 25°C , 2 cycles	—	design recommended value; actual production-line value requires tuning

4. Mass breakdown (per cell)

Formula caliber: $\text{mass} = \text{layer thickness} \times (1 - \text{porosity}) \times \text{density} \times \text{area}$ (contract formula, electrolyte excluded — parameter set has no electrolyte density); electrolyte row added with literature density 1.2 g/cm^3 ('estimate').

Component	Mass (g)	Source
Positive electrode coating	16.21	'r4_v4b_energy.json:layer_kg_m2.positive_electrode' $\times 0.1027 \text{ m}^2$
Negative electrode coating	11.11	'layer_kg_m2.negative_electrode' $\times 0.1027 \text{ m}^2$
Separator	0.18	'layer_kg_m2.separator' $\times 0.1027 \text{ m}^2$
Positive current collector (Al)	2.22	'layer_kg_m2.positive_cc' $\times 0.1027 \text{ m}^2$
Negative current collector (Cu)	5.52	'layer_kg_m2.negative_cc' $\times 0.1027 \text{ m}^2$
Subtotal (contract caliber, electrolyte excluded)	35.24	'r4_v4b_energy.json:mass_kg' (0.0352409 kg)
Electrolyte (literature density estimate)	7.69	§ 3 fill amount
Total incl. electrolyte	42.93	subtotal + electrolyte
Enclosure / tabs	Not modeled	—

Energy density: contract caliber 583.2 Wh/kg ('r4_v4b_energy.json:energy_density_wh_kg'); incl. electrolyte estimate $20.553 \text{ Wh} / 0.04293 \text{ kg} = 478.7 \text{ Wh/kg}$ ('inferred', formula noted).

5. Performance verification (V4b vs entry-0 criteria)

Item	Value	Criterion	Verdict	Source
Energy density	583.2 Wh/kg	≥ 392.61 Wh/kg	✓ pass	`cell/r4_v4b_energy.json:energy_density_wh_kg`
1C discharge capacity	5.6997 Ah	$\geq$ nominal 5.0 Ah	✓ pass	`cell/r4_v4b_1c_spme.json:capacity_ah`
4C fast-charge max temperature	326.46 K (53.31 ° C)	≤ 333.15 K	✓ pass (margin 6.69 K)	`cell/r4_v4b_4c_dfn.json:T_max_K` (326.459 K)
4C lithium plating	anode potential min +0.0215 V $\rightarrow$ plated = false	plated == false	✓ pass	`cell/r4_v4b_4c_dfn.json:anode_potential_v` min (mechanical)
4C CC acceptance	4.53 Ah = 79% of 5.70 Ah	(informational)	—	`inferred`: (t_end - t_vmin) $\times$ 20 A / 3600 = 815 s $\times$ 20 A / 3600; `capacity_ah` key unreliable for charge protocols (PyBaMM cycle-split artifact, recorded in evaluate R1 note)
Overcharge to 4.7 V (0.5C, 4.2+0.5 V protocol) max temperature	299.43 K	no thermal runaway	✓ pass	`cell/r4_v4b_overcharge.json:T_max_K`
Thermal runaway after overcharge	triggered = false	triggered == false	✓ pass	`cell/r4_v4b_tr.json:triggered` (mcp = 31.72 J/K = mass $\times$ 900; hA = 0.3186 = 60 $\times$ 0.00531 design-consistent)

Mechanical verdicts logged via `bda log-evaluate` round 4 (main + safety entries, both pass). Margin note: 4C T_max margin 6.69 K and anode margin +21.5 mV are the best combined margins across all four rounds.

6. Design notes (what was changed and why)

Baseline (Chen2020, R1) met energy density (400.3 Wh/kg) but failed 4C: T_max 354.29 K and anode -0.192 V (plating), acceptance 0.88 Ah $\rightarrow$ binding constraint = 4C polarization at cell scale $\rightarrow$ fallback to Stage 3 (architecture + thermal), no Stage 2 escalation (ED ceiling ≈ 550 Wh/kg assessment, R1).

- **R2 (kinetics + transport)** — plating fixed by: fine particles (positive 2.0 μ m / negative 1.5 – 2.5 μ m), electrolyte bridge σ 1.8 S/m / D 6e-10 / t 0.4 ('estimate'), negative porosity 0.32; geometry alone insufficient (V2c still plated, evaluate R2).
- **R3 (thermal)** — T_max fixed by liquid cooling h = 45 W/m² · K (V3b/V3c pass all criteria; h = 30 fails by 2.0 K, evaluate R3).
- **R4 (robustness)** — h = 60 W/m² · K (cold-plate class), negative thickness 96 μ m (N/P ≈ 1.35), negative particle 1.5 μ m (surface $\times 1.33$) to widen both margins (evaluate R4).

Transparency notes:

- The `Cell volume [m3]` override (2.514e-5) was scaled from the pre-collector-thinning stack (208.6 μ m); the as-designed stack is 194.6 μ m, so the thermal simulation volume is $\sim 7\%$ larger than the as-built stack — conservative direction for T_max (larger volume $\rightarrow$ larger lumped heat input at equal heat flux).
- Electrolyte σ / D / t are literature-based bridge estimates, not simulation outputs; they are marked `estimate` in the audit log and are the parameters a Stage 2 formulation round or true MD endorsement would verify (skipped: entry-0 meta `real\_compute: false`, endorse entry records the skip honestly).
- `capacity\_ah` in charge/overcharge protocol outputs is unreliable (PyBaMM cycle-split artifact); 4C acceptance derived mechanically from the voltage curve (formula in § 5).

All conclusion-grade values above are tool-output values ('sourced') or mechanically derived from them ('inferred'); no fabricated values.